

Chemistry Learner Assistant Bot

An AI-powered chemistry tutor that explains concepts, solves numericals, balances equations, and preps students for exams — built and prompt-engineered end-to-end on the Zapier Chatbots platform.

ROLE

AI Prompt & Conversation Designer

DOMAIN BACKGROUND

BS Chemistry · Classroom Teacher

PLATFORM

Zapier Chatbots (No-Code AI)

AUDIENCE

Chemistry Students, Grades 9–12 & Intro College

→ TRY THE LIVE BOT

01 Overview

The Problem

Chemistry is one of the subjects students most often abandon mid-way — dense vocabulary, multi-step numericals, and equation balancing all compound quickly. Most generic AI chatbots either oversimplify the chemistry or bury students in jargon, and neither approach actually helps a student walk into an exam prepared.

The Approach

I designed the bot the way I'd plan a lesson: a clear learning objective, a consistent explanatory structure, and a tone calibrated for a real classroom — patient, encouraging, and precise. Every instruction in the bot's directive was written and tested the way I'd write a rubric: specific enough to be reliable, flexible enough to meet a student wherever they are.

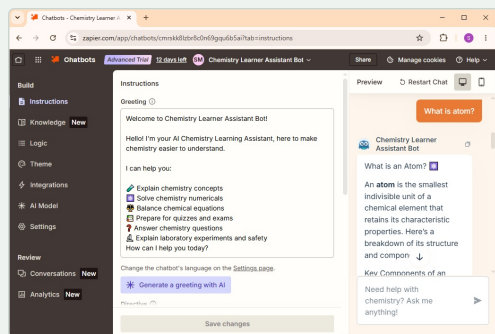
"Explain chemistry concepts in simple, easy-to-understand language" isn't a throwaway line in the prompt — it's the same standard I hold myself to in front of a whiteboard.

02 Anatomy of the Build

Six configuration layers, assembled like elements on a bench — each one doing a distinct job in how the bot thinks, knows, and presents itself.

In 01 INSTRUCTIONS Core directive defining objective, teaching style, and formatting rules (headings, steps, bullet points).	Kb 02 KNOWLEDGE Reference base of chemistry content the bot draws on for accurate, curriculum-aligned answers.	Lg 03 LOGIC Conversation rules that route a question to the right kind of response — concept, calculation, or safety.
Th 04 THEME Owl-tutor avatar, logo, and embed styling — a visual identity that reads as approachable, not clinical.	Ig 05 INTEGRATIONS Connection points for deployment — embeddable on a website, LMS, or tutoring portal.	Md 06 AI MODEL Underlying model configuration tuned for accuracy on numerical and equation-based chemistry.

03 Inside the Build



ONBOARDING & FIRST RESPONSE

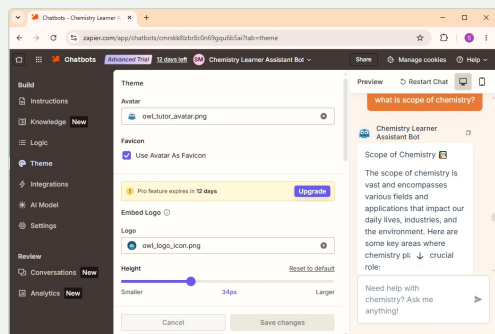
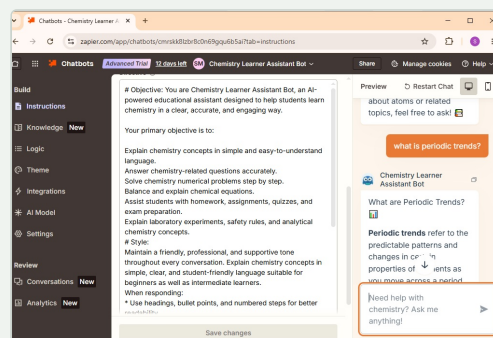
A warm, scoped greeting

The welcome message sets expectations up front — six concrete things the bot can do — so a student never has to guess what to ask. The sample exchange shows the bot handling “What is an atom?” with a structured, student-friendly breakdown rather than a dense wall of text.

PROMPT ENGINEERING

The directive that shapes every answer

This is the instruction layer I authored: a defined objective, six explicit responsibilities (explaining concepts, solving numericals, balancing equations, exam prep, lab safety), and formatting rules that keep every response scannable — headings, bullets, and numbered steps by default.



BRANDING & VISUAL IDENTITY

A friendly face for a technical subject

A custom owl-tutor avatar and matching logo soften the subject matter, while the embed sizing and favicon settings were tuned so the bot feels native wherever it's placed — a tutoring site, an LMS, or a landing page.

04 What the Bot Actually Does

- ✓ Explains chemistry concepts in plain, student-level language
- ✓ Balances chemical equations on request
- ✓ Supports quiz and exam preparation
- ✓ Solves numerical problems with visible step-by-step working
- ✓ Answers open-ended chemistry questions accurately
- ✓ Covers lab experiments and safety guidance

05 Skills Demonstrated

This project sits at the intersection of subject-matter expertise and applied AI product work — the two areas I now offer clients:

- AI Prompt Engineering
- Conversational UX Design
- Instructional Content Design
- No-Code AI Tools (Zapier)
- Chatbot Persona & Branding
- Subject-Matter Accuracy Review
- STEM Curriculum Alignment

Need a subject-accurate AI tutor or chatbot built for your students or brand?

I design and prompt-engineer AI assistants that are genuinely accurate in the subject matter — not just fluent-sounding. Available for chatbot builds, prompt engineering, and AI-assisted curriculum projects.

LIVE DEMO

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